trend-following portfolio, as either a positive (winner) or a negative (loser) signal is assigned to each instrument at each point in time.

4.1. Sub-portfolio Sign Analysis

Sign analysis allows us to see which sub-portfolios exhibit strong time series reversal. Here, we calculate the frequency of positive signs for sub-portfolio components during the post trend-following holding period (13–60 months). We divide the post holding period into four separate periods, each lasting for one year. These are 13–24 months, 25–36 months, 37–48 months, and 49–60 months after portfolio formation.

Based on the above-mentioned settings, (j, h) sub-portfolios are established for analysis. For instance, in a $(12, 12)$ scheme, the trend-following signal (winner/loser) at time t is based on the returns during months $t - 11$ to t for each individual instrument. Then, we determine whether the profits of these winners and losers are realised or not during the next $h = 12$ periods, i.e., $t + 1$ to $t + 12$. Finally, we calculate how many instruments in each subgroup generate positive returns over months $t + 13$ to $t + 60$ and report the rate of positive returns. The same process is followed for each month from January 1985 to March 2015.

Our primary focus is on the results during months 13–24 of the post holding period, when time series reversal occurs. Figure 5 shows these results as a plot of the aggregated probability of positive signs for all the instruments classified as one of the four sub-portfolios.[17] The loser subgroups, i.e., the realised losers and the contrarian losers, exhibit positive rates of around 65–66% and 63%, respectively, during this period. These success rates are much higher than the unconditional positive rate of 58.7%, which was calculated using all the 55 individual instruments from January 1985 to March 2015. We also employ a proportion test developed by Newcombe (1998a,b) to examine whether the success rates of these subgroups are statistically different from those for the unconditional rate. The results suggest that the rates of both the realised loser and contrarian loser sub-portfolios are significantly different

[17]We also show results of sign analysis for the rest of the post holding period (25–60 months) using different (j, h) settings. These results are reported in Section E of the Online Appendix.